ASHOK H

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PROFESSIONAL SUMMARY

Proactive and detail-oriented Software Engineer with hands-on enterprise experience in developing responsive web applications and scalable microservices. Proficient in **Angular 20**, **TypeScript**, **Python (Flask)**, and **Azure cloud infrastructure** (AKS, Cosmos DB). Proven ability in building real-time IoT monitoring dashboards, processing telemetry data, and optimizing frontend performance, with strong problem-solving skills across the full software development lifecycle, from UI design to containerized CI/CD deployments.

WORK EXPERIENCE

Software Engineer | Full-time — Feb 2026 - Present

ConnectSecure — *On-site • Bengaluru, Karnataka*

- **Architected scalable real-time monitoring dashboards** using **Angular 20, PrimeNG, and ApexCharts** to track City Gas Distribution (CGD) operations, visualizing live CNG dispenser sales, line pressures, and mass flow meter (MFM) telemetry across hundreds of BPCL Retail Outlets (ROs).
- **Integrated WebSockets and optimized Angular change-detection**, enabling live geospatial alert streaming while eliminating UI thread blocking and screen freeze during high-frequency telemetry bursts and modal dialog interactions.
- **Implemented IoT router and telemetry rule engine microservices in Python**, ingesting raw Modbus register payloads, accurately resolving Retail Outlet IDs (ROIDs), and automating safety threshold and emergency shutdown alerts.
- **Developed the Station-level Gas Reconciliation & Custody Transfer Billing Engine**, automating multi-unit volume conversions (kg, SCM, MMBTU), flagging custody imbalance penalty breaches (≤90% / ≥110%), and generating provisional billing PDF statements.
- **Designed dynamic administrative tooling**, including high-performance searchable autocomplete filters for retail outlets, custom date-range analytical queries, and interactive Modbus register mapping interfaces for rapid device onboarding.
- **Collaborated on containerized CI/CD deployments** using **Docker and Azure DevOps pipelines**, ensuring automated, zero-downtime releases across multi-cluster **Azure Kubernetes Service (AKS)** environments (DEV, QA, PROD).

Technologies / Skills Used : *Angular 20, TypeScript, Python (Flask), Azure Kubernetes Service (AKS), Azure Cosmos DB, WebSockets (Flask-SocketIO), ApexCharts, PrimeNG, Microservices Architecture, Docker, Azure DevOps (CI/CD)*

EDUCATION

University of Visvesvaraya College of Engineering (UVCE) | Bengaluru, India — 2022 - 2026

B.Tech – Electronics & Communication Engineering — *CGPA : 8.76*

PROJECTS

Conversational AI Agent – Agentic RAG System

Developer

- Built a system that takes a goal (answer a question about a PDF), breaks it into steps (retrieve relevant chunks, reason over context, synthesize answer), and recovers gracefully when a retrieval step returns poor results.
- Designed a **LangGraph agentic pipeline** with conditional routing, state management, and **5 decoupled nodes** – ingestion, chunking, embedding, retrieval, and generation – each independently testable and replaceable.
- Implemented document chunking and HuggingFace embedding generation; stored vectors in Qdrant for semantic similarity search, achieving **sub-200ms retrieval latency** on **50+ indexed documents**.
- Connected Gemini API for grounded response generation with prompt engineering and output validation; added speech recognition and synthesis for full voice I/O workflow.
- Built automated evaluation workflows covering **20+ test scenarios**; measured retrieval accuracy and response quality against benchmark sets, achieving **95%+ accuracy** – iterated prompts based on results.
- Shipped a working end-to-end agentic application with observable pipeline stages, structured error handling, and measurable quality metrics, including **AI products** and **data-heavy backends**.

Technologies / Tools Used : *Python, LangChain, LangGraph, Qdrant, HuggingFace, Gemini API*

Smart Agriculture ML System

Developer

- Built and deployed **Python ML models** on GCP for predictive analytics on real agricultural sensor data, achieving **85%+ prediction accuracy** on held-out validation sets.
- Designed the full data pipeline: ESP32 sensor collection → REST API ingestion → Firebase time-series store → GCP inference → mobile delivery, processing **100+ readings/minute** with zero data loss.
- Compared model approaches and tuned for accuracy vs. latency trade-offs; integrated model outputs via REST API with **sub-2 second end-to-end latency** under live competition load.

Technologies / Tools Used : *Python, GCP, Firebase, ESP32*

Car Rental Platform

Developer

- Designed a modular **REST API backend** with **12+ endpoints**, JWT authentication, role-based access control, and structured error handling and logging throughout – built to be readable and maintainable by other engineers.
- Managed codebase with **Git** across structured feature branches; containerised with **Docker**; maintained full technical documentation – **zero defects post-deployment**.

Technologies / Tools Used : *Python, Node.js, Express.js, React.js, MongoDB, Docker*

SKILLS

Databases :	Azure Cosmos DB, Qdrant, MongoDB, MySQL, Firebase
Frameworks & Libraries :	LangChain, LangGraph, Node.js, Express, React, FastAPI, Tailwind CSS, Angular, Flask
Coursework :	Object Oriented Programming, Computer Networks, Data Structures & Algorithms, Operating Systems
Programming Languages :	Python, JavaScript, C, C++, Java, SQL
Soft Skills :	Teamwork, Communication, Decision Making, Problem Solving, Adaptability
Tools & Platforms :	GCP, Azure IoT, Git, Docker, Linux, Postman, GitHub, CI/CD, Kubernetes

AWARDS & ACHIEVEMENTS

- Shipped an end-to-end AI system with real sensor data and deployed ML models under competition conditions.